

Course Registration System – Final Report

Blockchain and Smart Contract Development

Project by Ghazal E Ashar & Shahzeb Ahmed Iqbal

Submitted to Ms Noor ul Ain Afaq

Summary

This project implements a complete course registration system using blockchain technology, featuring automated token economics, real-time transaction monitoring, and advanced security patterns and a responsive frontend.

Key Features & Functionality

Student Portal

- **Course Registration:** Browse and register for available courses
- **Token Requests:** Request CRST tokens by paying ETH upfront
- **Payment Management:** View registration status and fee payment history

Admin Portal

- **Course Management:** Create, update, activate/deactivate courses
- **Token Request Approval:** Review and approve/reject student token requests
- **Financial Operations:** Withdraw ETH/CRST funds with automatic token burning

Development & Deployment

Smart Contract Layer

- **CRSTToken.sol:** ERC20 token with dual auto-burn mechanics.
- **CourseRegistration.sol:** Main business logic contract with role-based access control.
- **Gas Optimizations:** Storage packing, batch operations, immutable variables.
- **Security Features:** ReentrancyGuard, input validation, role based controls.

Frontend Integration

- **Technology Stack:** JavaScript, Ethers.js, Bootstrap, MetaMask integration.
- **Real-time Updates:** Event-driven UI with automatic transaction monitoring.
- **User Experience:** Wallet-only authentication.

Deployment:

- **Framework:** Hardhat for smart contract development and testing
- **Network:** Local Ethereum blockchain (localhost:8545, chainId: 31337)
- **Configuration:** Automatic ABI and address generation for seamless frontend integration

Innovation & Differentiation

- **Pre-paid Token Requests:** Students pay ETH upfront, eliminating credit risk
- **Economic Stability:** Automated supply management maintains token value and backing ratio with dual auto-burn system without manual intervention

Conclusion

This course registration system successfully demonstrates advanced blockchain development capabilities, combining innovative tokenomics with practical user experience design. The dual auto-burn mechanism represents a novel approach to automated monetary policy, while the comprehensive security implementation and gas optimizations showcase production-ready smart contract development skills.